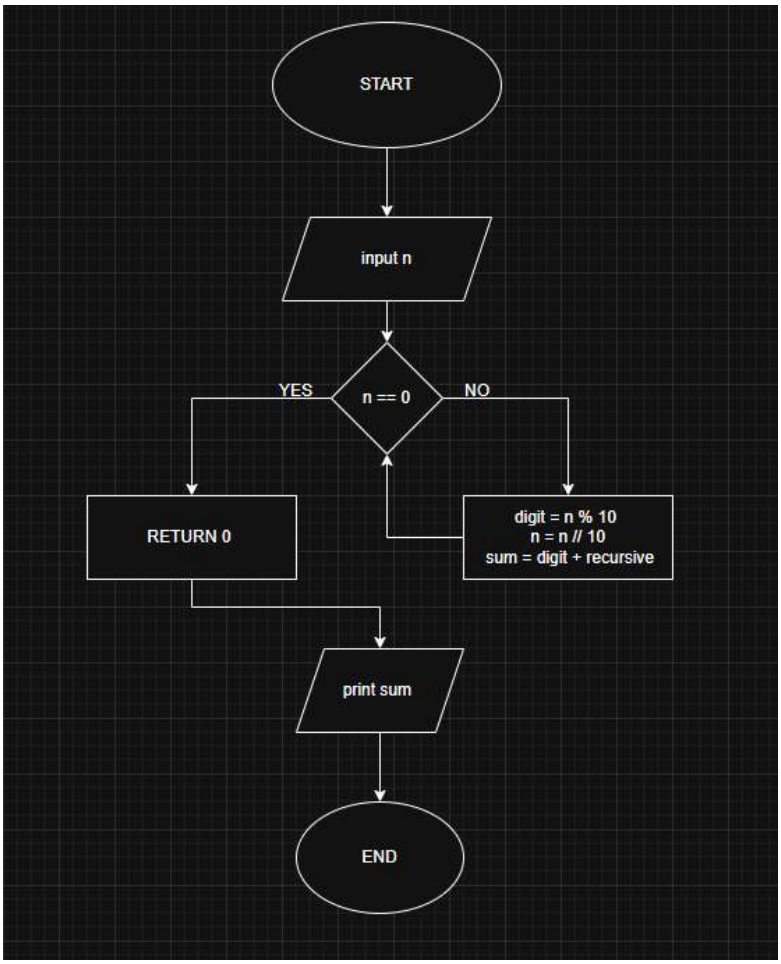
11.1.2 SUM OF DIGIT USING RECURSION

ALGORITHM

Step 1: Start
Step 2: Input integer `n`
Step 3: Check if `n == 0` If yes → return `0`
Step 4: Otherwise
Compute `(n % 10)` → last digit
Compute `sum_of_digits_recursive(n // 10)` → recursive call
Step 5: Return `(n % 10) + sum_of_digits_recursive(n // 10)`
Step 6: Print result
Step 7: End

FLOWCHART



PROGRAM

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11.1.2. Sum of Digits using Recursion 01:22

Write a recursive function **sum_of_digits_recursive(n)** that calculates the sum of the digits of a given non-negative integer n.

Input Format:

- The input consists of a single non-negative integer n.

Output Format:

- The output should be the sum of the digits of the non-negative integer n.

Constraints:

- $0 \leq n \leq 10^9$

Sample Test Cases +

Explorer

SumofDi...

Submit

```
"""def sum_of_digits_recursive(n):
    
    # Write your code here
    
    return    sum_of_digits_recursive(  )



# Write your code here

result = sum_of_digits_recursive(number)
"""
def sum_of_digits_recursive(n):
    if n == 0:
        return 0
    else:
```

Debugger

Average time 0.003 s 3.40 ms

Maximum time 0.007 s 7.00 ms

2 out of 2 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 7 ms Debug

Expected output	Actual output
23	23
5	5

Test case 2 4 ms

Terminal Test cases